

Rand

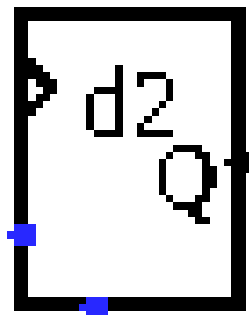
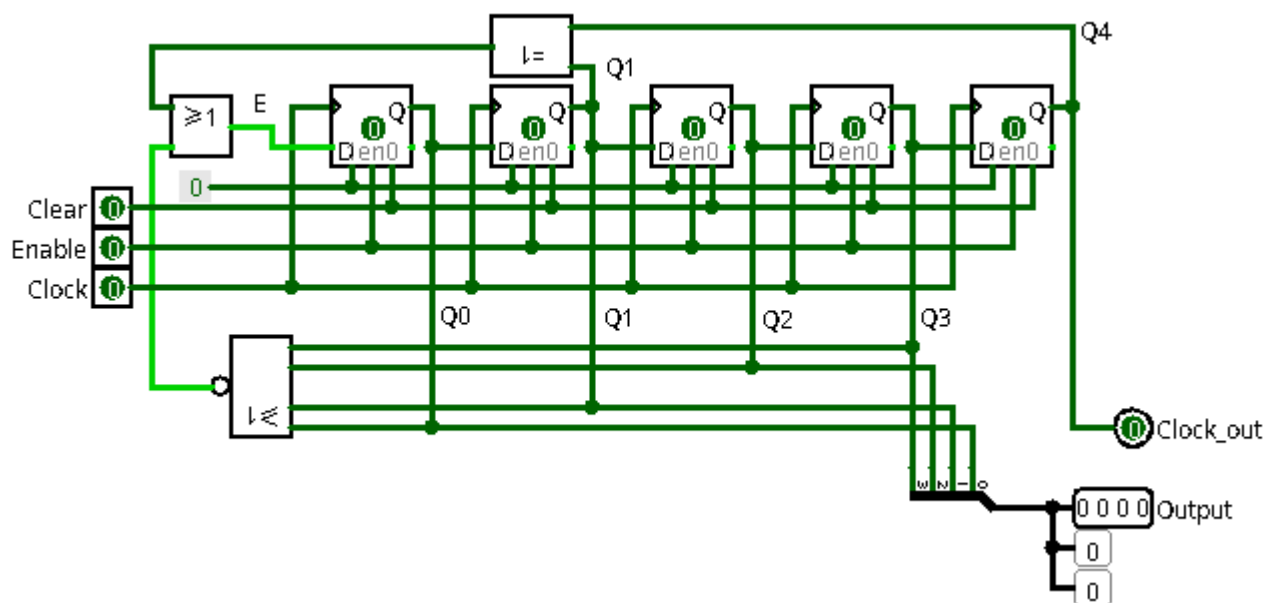
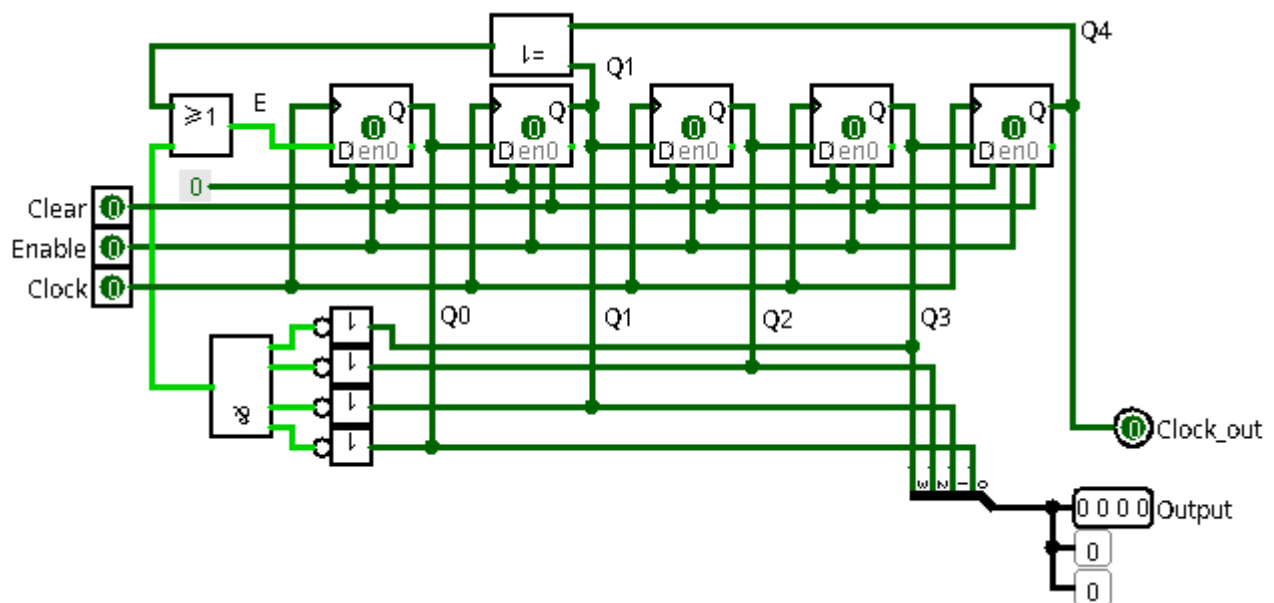


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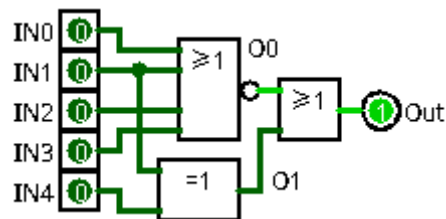
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1) rand (semi-aléatoire): (SRAND4 - SRAND4M)



2) Coupé en 2 parties :

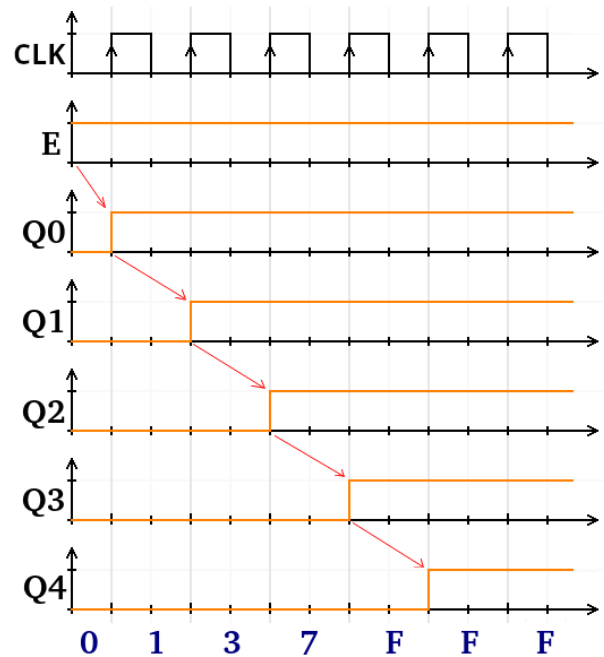
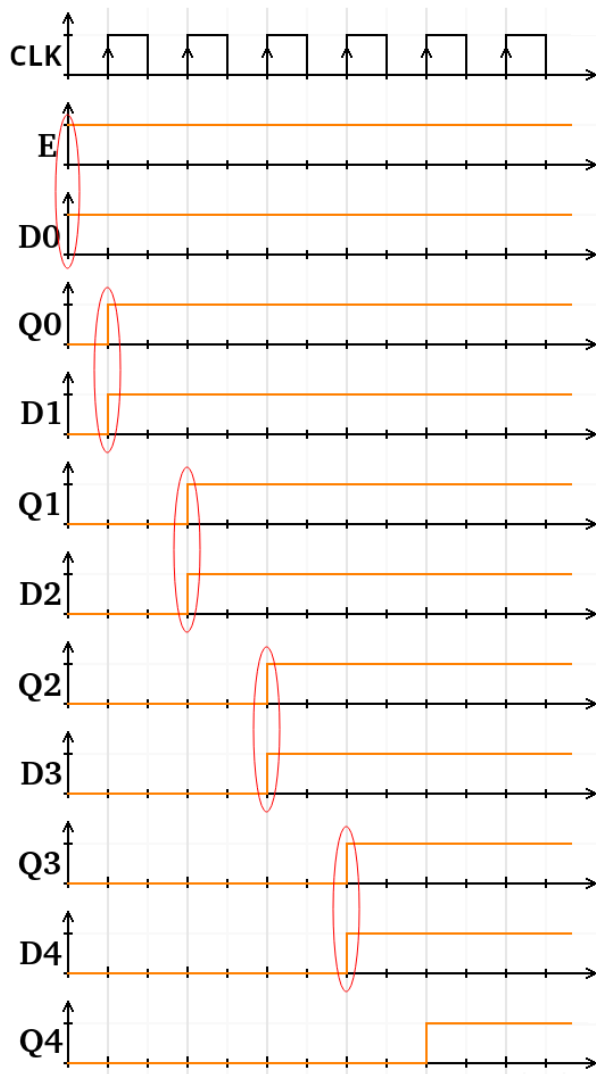
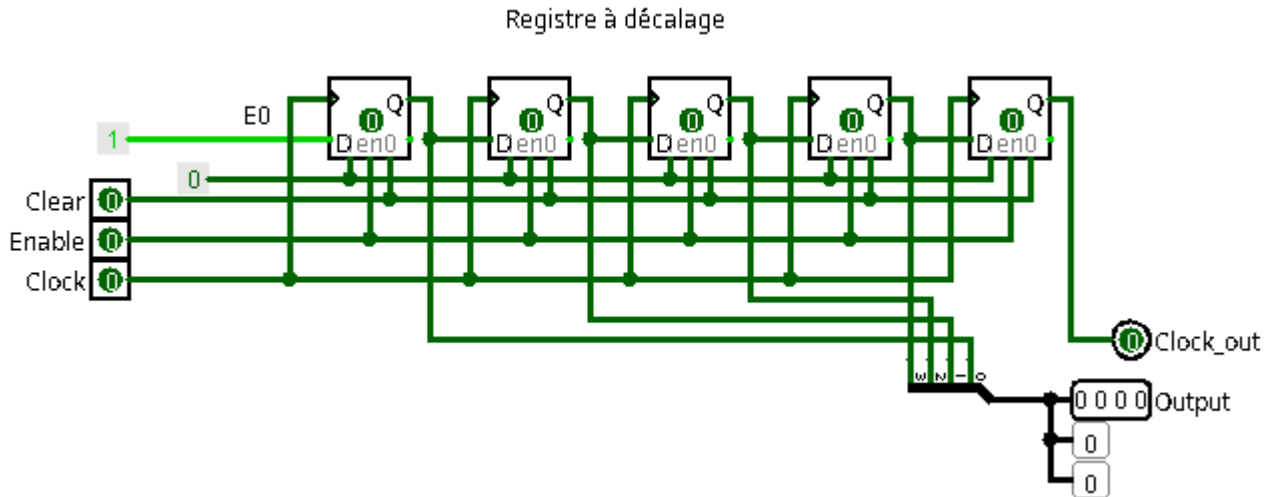
2-1) Changement d'état de l'entrée : (CSTATE)

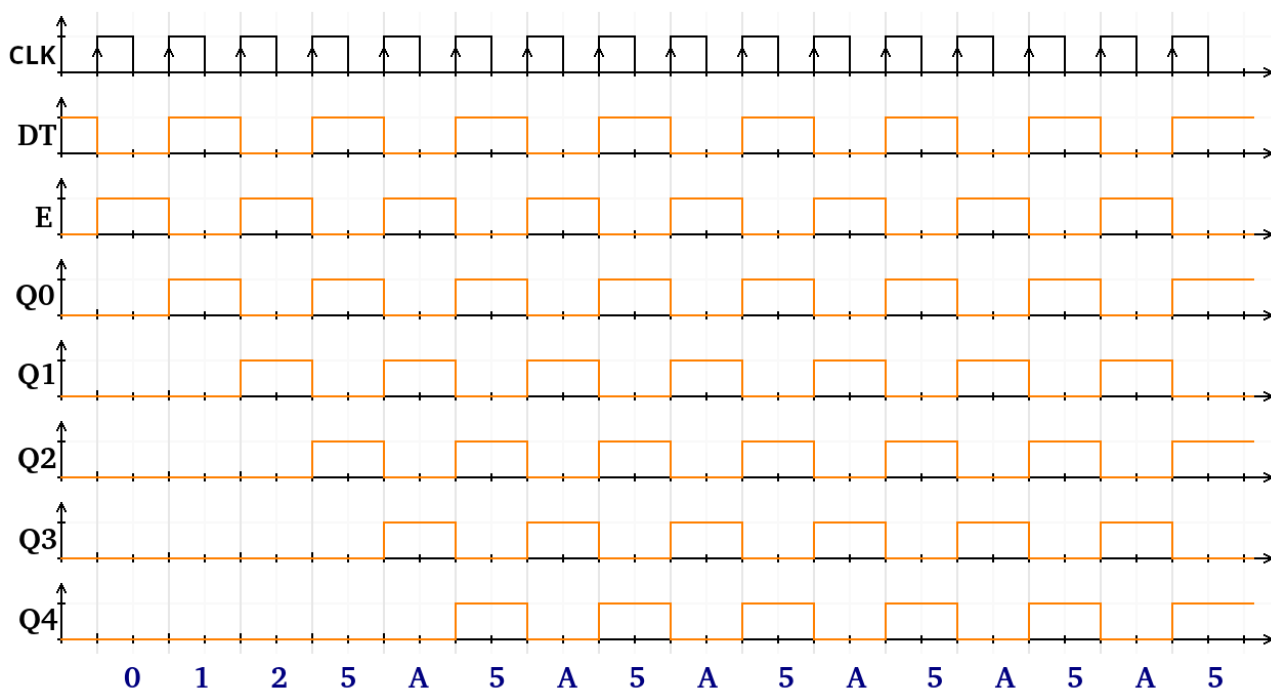
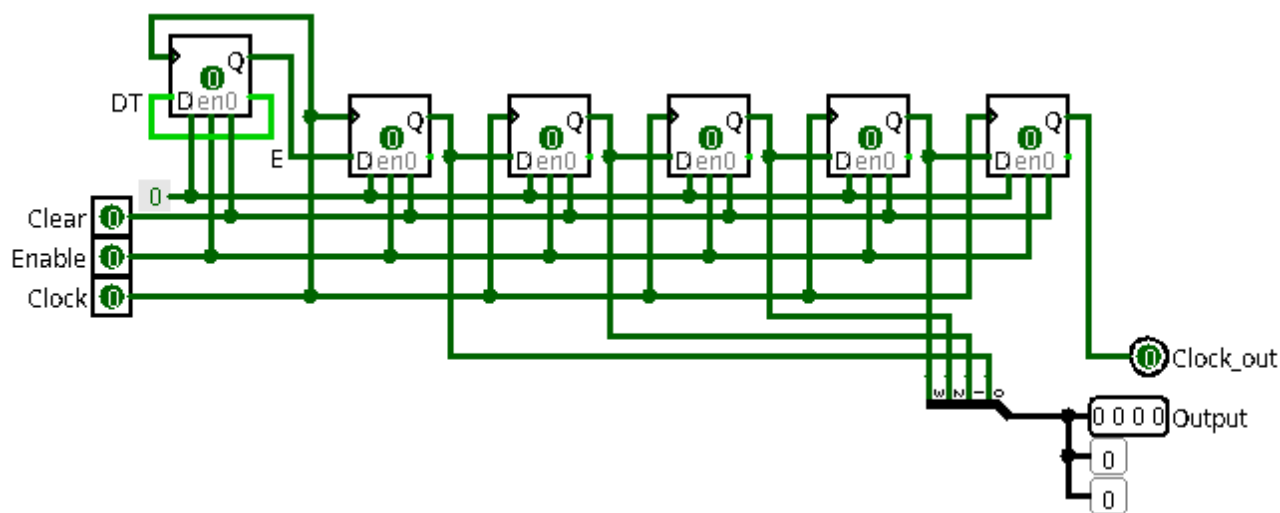


IN4	IN3	IN2	IN1	IN0	O0	O1	OUT
0	0	0	0	0	1	0	1
0	0	0	0	1	0	0	0
0	0	0	1	0	0	1	1
0	0	0	1	1	0	1	1
0	0	1	0	0	0	0	0
0	0	1	0	1	0	0	0
0	0	1	1	0	0	1	1
0	0	1	1	1	0	1	1
0	1	0	0	0	0	0	0
0	1	0	0	1	0	0	0
0	1	0	1	0	0	1	1
0	1	0	1	1	0	1	1
0	1	1	0	0	0	0	0
0	1	1	0	1	0	0	0
0	1	1	1	0	0	1	1
0	1	1	1	1	0	1	1
1	0	0	0	0	0	1	1
1	0	0	0	1	0	1	1
1	0	0	1	0	0	0	0
1	0	0	1	1	0	0	0
1	0	1	0	0	0	1	1
1	0	1	0	1	0	1	1
1	0	1	1	0	0	0	0
1	0	1	1	1	0	0	0
1	1	0	0	0	0	1	1
1	1	0	0	1	0	1	1
1	1	0	1	0	0	0	0
1	1	0	1	1	0	0	0
1	1	1	0	0	0	1	1
1	1	1	0	1	0	1	1
1	1	1	1	0	0	0	0
1	1	1	1	1	0	0	0

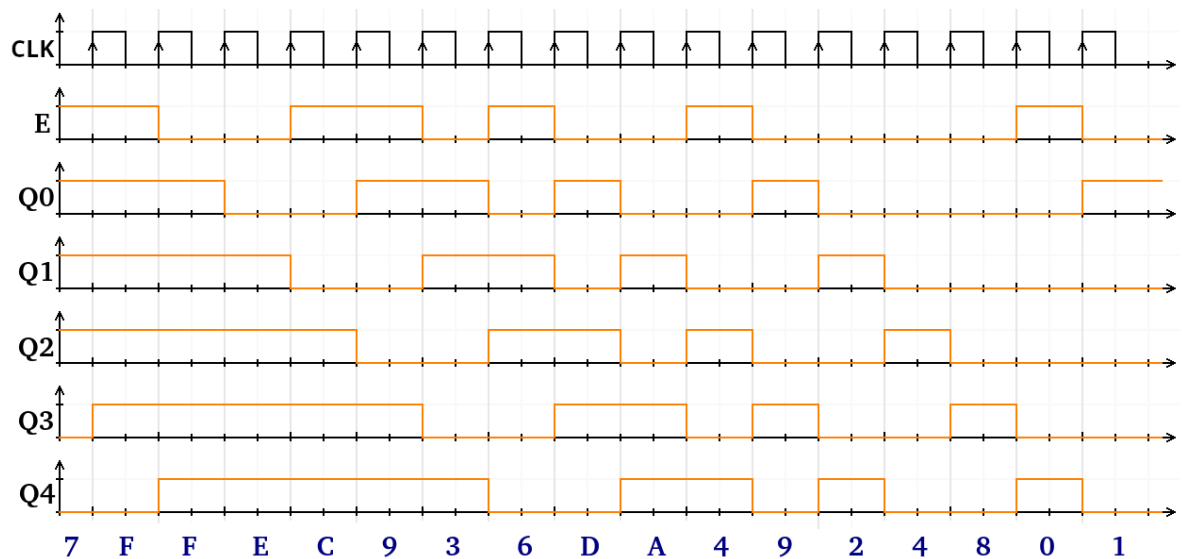
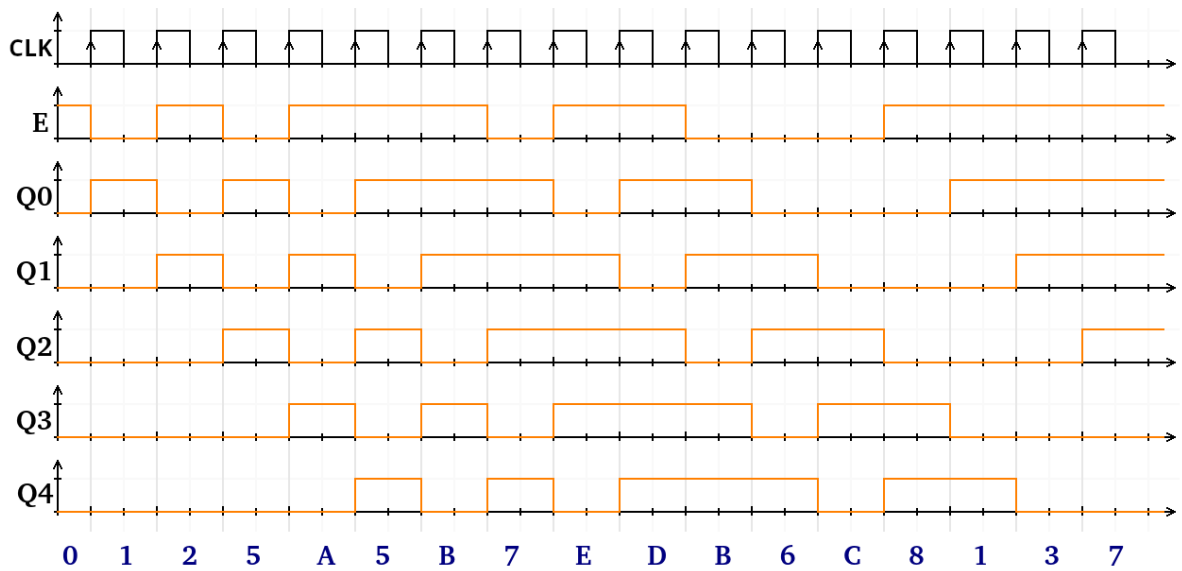
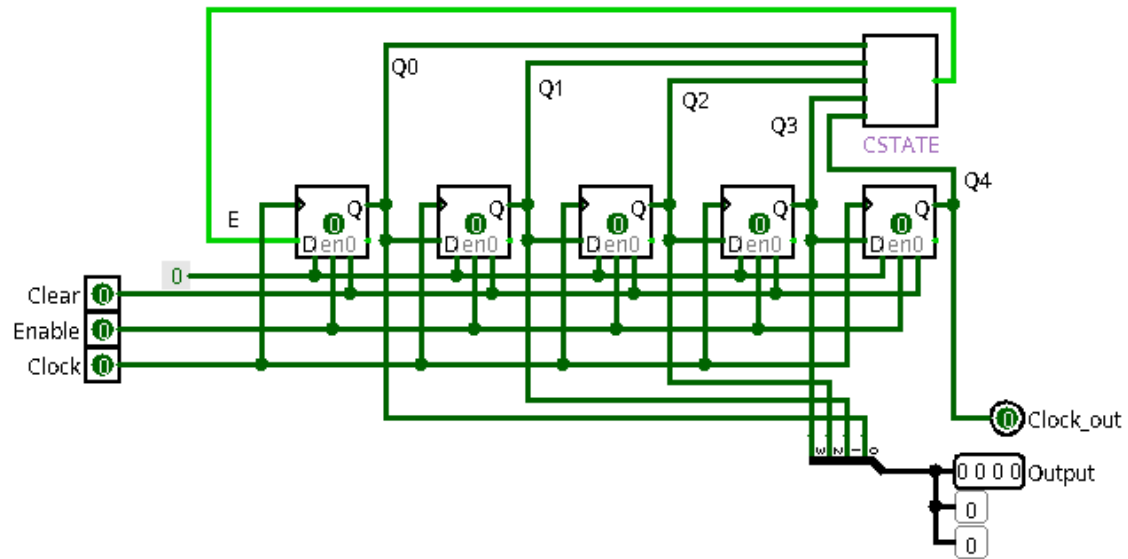
IN4	IN1	O1
0	0	0
0	0	0
0	1	1
0	1	1
0	0	0
0	0	0
0	1	1
0	1	1
0	0	0
0	0	0
0	1	1
0	1	1
1	0	1
1	0	1
1	1	0
1	1	0
1	0	1
1	0	1
1	1	0
1	1	0
1	0	1
1	0	1
1	1	0
1	1	0

2-2) Registre à décalage série : (SISO - SISO2)



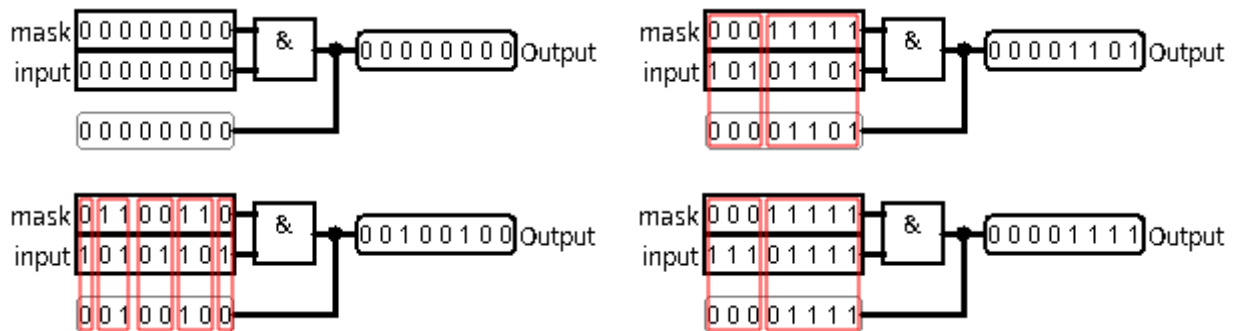


2-3) En totalité : (SRAND4S)

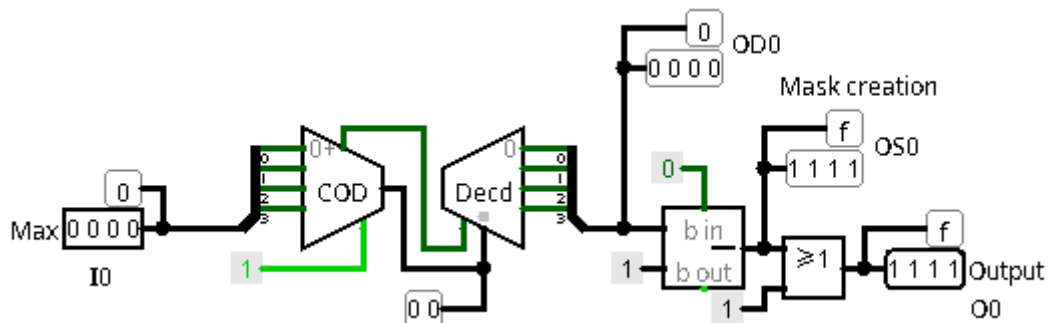


3) Avec valeur maximale :

3-1) C'est quoi un masque : (MASK)

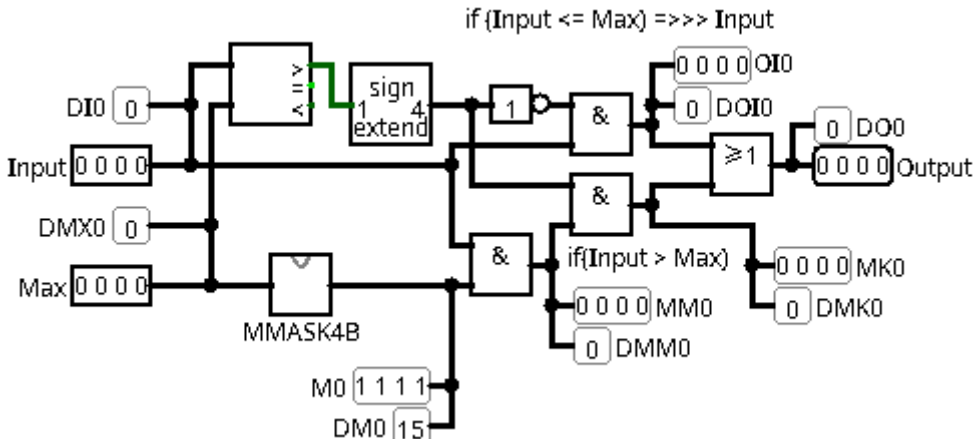


3-2) Création d'un masque pour le nombre aléatoire : (MMASK4B)



i0					i0V	OD0					OD0V	OS0					OS0V	OO					OOV
0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	15	1	1	1	1	15			
0	0	0	1	1	0	0	0	0	1	1	0	0	0	0	0	0	0	0	1	1			
0	0	1	0	2	0	0	1	0	2	2	0	0	0	1	1	0	0	0	1	1			
0	0	1	1	3	0	0	1	0	2	2	0	0	0	1	1	0	0	0	1	1			
0	1	0	0	4	0	1	0	0	4	4	0	0	1	1	3	0	0	1	1	3			
0	1	0	1	5	0	1	0	0	4	4	0	0	1	1	3	0	0	1	1	3			
0	1	1	0	6	0	1	0	0	4	4	0	0	1	1	3	0	0	1	1	3			
0	1	1	1	7	0	1	0	0	4	4	0	0	1	1	3	0	0	1	1	3			
1	0	0	0	8	1	0	0	0	8	8	0	1	1	1	7	0	1	1	1	7			
1	0	0	1	9	1	0	0	0	8	8	0	1	1	1	7	0	1	1	1	7			
1	0	1	0	10	1	0	0	0	8	8	0	1	1	1	7	0	1	1	1	7			
1	0	1	1	11	1	0	0	0	8	8	0	1	1	1	7	0	1	1	1	7			
1	1	0	0	12	1	0	0	0	8	8	0	1	1	1	7	0	1	1	1	7			
1	1	0	1	13	1	0	0	0	8	8	0	1	1	1	7	0	1	1	1	7			
1	1	1	0	14	1	0	0	0	8	8	0	1	1	1	7	0	1	1	1	7			
1	1	1	1	15	1	0	0	0	8	8	0	1	1	1	7	0	1	1	1	7			

3-3) Valeur max : (RVM4B)



DMMO							DMMO							DMMO						
DMX0	DIO	DM0	DMM0	DMK0	DOIO	DO0	DMX0	DIO	DM0	DMM0	DMK0	DOIO	DO0	DMX0	DIO	DM0	DMM0	DMK0	DOIO	DO0
0	0	15	0	0	0	0	1	0	1	0	0	0	0	2	0	1	0	0	0	0
0	1	15	1	0	1	1	0	1	1	1	0	0	1	3	0	1	1	0	0	1
0	2	15	2	0	2	2	1	2	1	0	0	0	0	3	2	1	0	0	0	2
0	3	15	3	0	3	3	1	3	1	1	0	0	1	3	3	1	1	0	0	3
0	4	15	4	0	4	4	1	4	1	0	0	0	0	3	4	1	0	0	0	0
0	5	15	5	0	5	5	1	5	1	1	0	0	1	3	5	1	1	1	0	1
0	6	15	6	0	6	6	1	6	1	0	0	0	0	3	6	1	0	0	0	0
0	7	15	7	0	7	7	1	7	1	1	1	0	1	3	7	1	1	1	0	1
0	8	15	8	0	8	8	1	8	1	0	0	0	0	3	8	1	0	0	0	0
0	9	15	9	0	9	9	1	9	1	1	1	0	1	3	9	1	1	1	0	1
0	10	15	10	0	10	10	1	10	1	0	0	0	0	3	10	1	0	0	0	0
0	11	15	11	0	11	11	1	11	1	1	1	0	1	3	11	1	1	1	0	1
0	12	15	12	0	12	12	1	12	1	0	0	0	0	3	12	1	0	0	0	0
0	13	15	13	0	13	13	1	13	1	1	1	0	1	3	13	1	1	1	0	1
0	14	15	14	0	14	14	1	14	1	0	0	0	0	3	14	1	0	0	0	0
0	15	15	15	0	15	15	1	15	1	1	1	0	1	3	15	1	1	1	0	1

DMMO							DMMO							DMMO						
DMX0	DIO	DM0	DMM0	DMK0	DOIO	DO0	DMX0	DIO	DM0	DMM0	DMK0	DOIO	DO0	DMX0	DIO	DM0	DMM0	DMK0	DOIO	DO0
4	0	3	0	0	0	0	5	0	3	0	0	0	0	6	0	3	0	0	0	0
4	1	3	1	0	1	1	5	1	3	1	0	1	1	6	1	3	1	0	1	1
4	2	3	2	0	2	2	5	2	3	2	0	2	2	6	2	3	2	0	2	2
4	3	3	3	0	3	3	5	3	3	3	0	3	3	6	3	3	3	0	3	3
4	4	3	0	0	4	4	5	4	3	0	0	4	4	6	4	3	0	0	4	4
4	5	3	1	1	0	1	5	5	3	1	1	0	5	6	5	3	1	0	5	5
4	6	3	2	2	0	2	5	6	3	2	2	0	2	6	6	3	2	0	6	6
4	7	3	3	0	3	3	5	7	3	3	0	3	3	6	7	3	3	0	7	7
4	8	3	0	0	0	0	5	8	3	0	0	0	0	6	8	3	0	0	0	0
4	9	3	1	1	0	1	5	9	3	1	1	0	1	6	9	3	1	1	0	1
4	10	3	2	2	0	2	5	10	3	2	2	0	2	6	10	3	2	2	0	2
4	11	3	3	0	3	3	5	11	3	3	0	3	3	6	11	3	3	0	3	3
4	12	3	0	0	0	0	5	12	3	0	0	0	0	6	12	3	0	0	0	0
4	13	3	1	1	0	1	5	13	3	1	1	0	1	6	13	3	1	1	0	1
4	14	3	2	2	0	2	5	14	3	2	2	0	2	6	14	3	2	2	0	2
4	15	3	3	0	3	3	5	15	3	3	0	3	3	6	15	3	3	0	3	3

DMMO							DMMO							DMMO						
DMX0	DIO	DM0	DMM0	DMK0	DOIO	DO0	DMX0	DIO	DM0	DMM0	DMK0	DOIO	DO0	DMX0	DIO	DM0	DMM0	DMK0	DOIO	DO0
8	0	7	0	0	0	0	9	0	7	0	0	0	0	10	0	7	0	0	0	0
8	1	7	1	0	1	1	9	1	7	1	0	1	1	11	0	7	1	0	0	1
8	2	7	2	0	2	2	9	2	7	2	0	2	2	11	2	7	2	0	2	2
8	3	7	3	0	3	3	9	3	7	3	0	3	3	11	3	7	3	0	3	3
8	4	7	4	0	4	4	9	4	7	4	0	4	4	11	4	7	4	0	4	4
8	5	7	5	0	5	5	9	5	7	5	0	5	5	11	5	7	5	0	5	5
8	6	7	6	0	6	6	9	6	7	6	0	6	6	11	6	7	6	0	6	6
8	7	7	7	0	7	7	9	7	7	7	0	7	7	11	7	7	7	0	7	7
8	8	7	0	0	8	8	9	8	7	0	0	8	8	11	8	7	0	0	8	8
8	9	7	1	1	0	1	9	9	7	1	0	9	9	11	9	7	1	0	9	9
8	10	7	2	2	0	2	9	10	7	2	2	0	2	11	10	7	2	0	10	10
8	11	7	3	3	0	3	9	11	7	3	3	0	3	11	11	7	3	0	11	11
8	12	7	4	4	0	4	9	12	7	4	4	0	4	11	12	7	4	0	12	12
8	13	7	5	5	0	5	9	13	7	5	5	0	5	11	13	7	5	0	13	13
8	14	7	6	6	0	6	9	14	7	6	6	0	6	11	14	7	6	0	14	14
8	15	7	7	7	0	7	9	15	7	7	7	0	7	11	15	7	7	0	15	15

DMMO							DMMO							DMMO						
DMX0	DIO	DM0	DMM0	DMK0	DOIO	DO0	DMX0	DIO	DM0	DMM0	DMK0	DOIO	DO0	DMX0	DIO	DM0	DMM0	DMK0	DOIO	DO0
12	0	7	0	0	0	0	13	0	7	0	0	0	0	14	0	7	0	0	0	0
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12	2	7	2	0	2	2	13	2	7	2	0	2	2	15	2	7	2	0	2	2
12	3	7	3	0	3	3	13	3	7	3	0	3	3	15	3	7	3	0	3	3
12	4	7	4	0	4	4	13	4	7	4	0	4	4	15	4	7	4	0	4	4
12	5	7	5	0	5	5	13	5	7	5	0	5	5	15	5	7	5	0	5	5
12	6	7	6	0	6	6	13	6	7	6	0	6	6	15	6	7	6	0	6	6
12	7	7	7	0	7	7	13	7	7	7	0	7	7	15	7	7	7	0	7	7
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12	9	7	1	1	0	9	13	9	7	1	0	9	9	15	9	7	1	0	9	9
12	10	7	2	2	0	10	13	10	7	2	0	10	10	15	10	7	2	0	10	10
12	11	7	3	3	0	11	13	11	7	3	0	11	11	15	11	7	3	0	11	11
12	12	7	4	4	0	12	13	12	7	4	0	12	12	15	12	7	4	0	12	12
12	13	7	5	5	0	13	13	13	7	5	0	13	13	15	13	7	5	0	13	13
12	14	7	6	6	0	14	13	14	7	6	0	14	14	15	14	7	6	0	14	14
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D10	DMX0	DO0	D10	DMX0	DO0	D10	DMX0	DO0	D10	DMX0	DO0	D10	DMX0	DO0	D10	DMX0	DO0	D10	DMX0	DO0	D10	DMX0	DO0	D10	DMX0	DO0	D10	DMX0	DO0	
0	8	0	0	9	0	0	10	0	0	11	0	0	12	0	0	13	0	0	14	0	0	15	0	0	16	0	0	0	17	0
1	8	1	1	9	1	1	10	1	1	11	1	1	12	1	1	13	1	1	14	1	1	15	1	1	16	1	1	1	17	1
2	8	2	2	9	2	2	10	2	2	11	2	2	12	2	2	13	2	2	14	2	2	15	2	2	16	2	2	2	17	2
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5	8	5	5	9	5	5	10	5	5	11	5	5	12	5	5	13	5	5	14	5	5	15	5	5	16	5	5	5	17	5
6	8	6	6	9	6	6	10	6	6	11	6	6	12	6	6	13	6	6	14	6	6	15	6	6	16	6	6	6	17	6
7	8	7	7	9	7	7	10	7	7	11	7	7	12	7	7	13	7	7	14	7	7	15	7	7	16	7	7	7	17	7
8	8	8	8	9	8	8	10	8	8	11	8	8	12	8	8	13	8	8	14	8	8	15	8	8	16	8	8	8	17	8
9	8	1	9	9	9	9	10	9	9	11	9	9	12	9	9	13	9	9	14	9	9	15	9	9	16	9	9	9	17	9
10	8	2	10	9	2	10	10	10	10	11	10	10	12	10	10	13	10	10	14	10	10	15	10	10	16	10	10	10	17	10
11	8	3	11	9	3	11	10	3	11	11	11	11	12	11	11	13	11	11	14	11	11	15	11	11	16	11	11	11	17	11
12	8	4	12	9	4	12	10	4	12	11	4	12	12	12	12	13	12	12	14	12	12	15	12	12	16	12	12	12	17	12
13	8	5	13	9	5	13	10	5	13	11	5	13	12	5	13	13	13	13	14	13	13	15	13	13	16	13	13	13	17	13
14	8	6	14	9	6	14	10	6	14	11	6	14	12	6	14	13	6	14	14	14	14	15	14	14	16	14	14	14	17	14
15	8	7	15	9	7	15	10	7	15	11	7	15	12	7	15	13	7	15	14	7	15	15	15	15	16	15	15	15	17	15

